

- 2 Approximately 50% of cases of coronary heart disease are thought to be a result of high blood pressure. Approximately 25% of adults worldwide have high blood pressure.

Blood pressure changes during the cardiac cycle, so the measurement of blood pressure uses two values: systolic blood pressure and diastolic blood pressure.

A person who is described as having high blood pressure will have a resting systolic blood pressure of 140 mm Hg or higher and a resting diastolic blood pressure of 90 mm Hg or higher.

There is some evidence that eating garlic can have positive effects on human health, including benefits to the cardiovascular system.

The storage organ (bulb) of the garlic plant, *Allium sativum*, is eaten worldwide. Figure 2.1 shows a garlic bulb.



Figure 2.1

Garlic extract can be obtained from garlic bulbs and placed into capsules that can be swallowed.

Scientists investigated the effect of eating garlic capsules on the blood pressure of people that had been diagnosed with high blood pressure.

The investigation studied 84 men and women of various ages. Each person was randomly placed in one of four groups and asked to follow a specific treatment for 12 weeks.

- 21 people ate one garlic capsule each day.
- 21 people ate two garlic capsules each day.
- 21 people ate four garlic capsules each day.
- 21 people were placed in a control group.

All the people continued taking any medication for high blood pressure that they had previously been prescribed.

The resting systolic blood pressure of each person was measured at the start of the investigation and at intervals during the 12 weeks.

- (a) Identify the independent variable **and** the dependent variable in this investigation.

independent variable

dependent variable

[2]

- (b) Suggest a suitable treatment for the people in the control group.

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..... [1]

The standard errors (SE) for each value are shown in brackets.

number of weeks after the start of the investigation	mean resting systolic blood pressure/mm Hg (± SE)			
	control group	1 garlic capsule each day	2 garlic capsules each day	4 garlic capsules each day
0	148.8 (± 2.8)	148.7 (± 2.8)	149.3 (± 2.6)	149.4 (± 2.7)
4	142.8 (± 2.9)	137.7 (± 2.8)	144.5 (± 2.7)	138.3 (± 2.7)
8	140.0 (± 2.9)	135.2 (± 2.8)	131.4 (± 2.6)	132.6 (± 2.9)
12	138.1 (± 3.1)	138.8 (± 3.1)	126.8 (± 2.8)	132.4 (± 3.0)

State the type of **data** represented by the resting systolic blood pressure.

..... [1]

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[3]



- (iii) The scientists decided that the t -test was **not** an appropriate choice of statistical test for the investigation.

Suggest **two** reasons why the t -test is **not** an appropriate choice of statistical test for the investigation.

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- (d) Suggest **two** changes that could be made to the investigation method that may increase confidence in the conclusions.

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..... [2]

[Total: 11]